

Football Analysis Full Technical Report

Detailed project documentation generated from the local repository state.

Abstract

- This project analyzes football videos with a pipeline that detects objects, tracks them across frames, assigns teams, estimates motion, maps coordinates to a pitch, and renders analytics overlays.

Project Overview

- Main goal: transform raw football footage into structured visual analytics outputs.
- Core features: detection, tracking, team color clustering, camera motion estimation, perspective transform, ball assignment, speed and distance overlays.

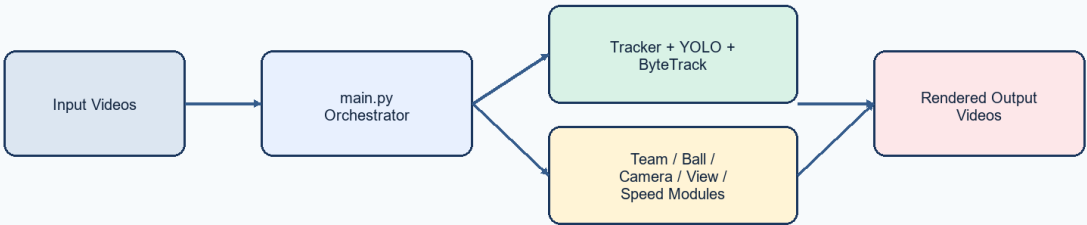
Runtime Configuration

- Input video path: input_videos/1.mp4
- resize_width: None
- max_frames: 1200
- Output save path expression: f'output_videos/{input_video_name}_output.avi'

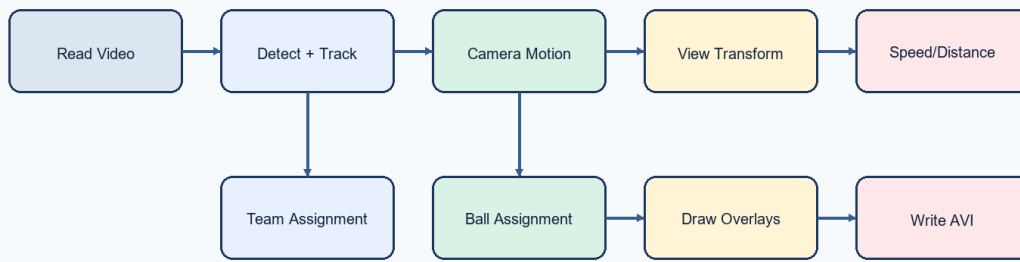
Dataset Summary

- Dataset URL: <https://universe.roboflow.com/roboflow-jvuqo/football-players-detection-3zvbc/dataset/1>
- Classes: ball, goalkeeper, player, referee
- Split counts: train=612, valid=38, test=13

System Architecture



Runtime Workflow



Important File Inventory

File	Purpose
main.py	Primary execution script that orchestrates the full inference, tracking, analytics, and rendering pipeline.
yolo_inference.py	Standalone quick inference script for directly testing YOLO predictions on a video.
README.md	Human-readable overview, setup instructions, and project summary.
requirements.txt	Python dependency list required to run the project.
trackers/tracker.py	Core tracking module that runs YOLO inference, structures tracks, interpolates ball detections, and draws tracks.
trackers/__init__.py	Package export file for Tracker.
team_assigner/team_assigner.py	Assigns players to one of two teams using KMeans clustering over jersey-color crops.
team_assigner/__init__.py	Package export file for TeamAssigner.
player_ball_assigner/player_ball_assigner.py	Chooses the player closest to the detected ball.
player_ball_assigner/__init__.py	Package export file for PlayerBallAssigner.
camera_movement_estimator/camera_movement_estimator.py	Computes camera motion using Lucas-Kanade optical flow on edge features and renders motion text.
camera_movement_estimator/__init__.py	Package export file for CameraMovementEstimator.
speed_and_distance_estimator/speed_and_distance_estimator.py	Computes and renders player speed and distance using transformed coordinates.
speed_and_distance_estimator/__init__.py	Package export file for SpeedAndDistanceEstimator.
view_transformer/view_transformer.py	Defines and applies a perspective transform from image coordinates to pitch coordinates.
view_transformer/__int__.py	Package export file for ViewTransformer; likely intended to be named __init__.py.
utils/bbox_utils.py	Geometry helper functions for centers, widths, distances, and player foot locations.
utils/video_utils.py	Video reading and writing helpers with optional resize and frame limiting.
utils/__init__.py	Package export file for utility functions.
training/football_training_yolo_v5.ipynb	Notebook describing dataset download, Ultralytics training, and weight export.
development_and_analysis/color_assignment.ipynb	Notebook used to prototype jersey-color clustering with KMeans.
training/football-players-detection-1/data.yaml	Ultralytics training configuration and class metadata file.
training/football-players-detection-1/README.datasetflow	Datasetflow dataset export metadata and augmentation notes.
training/football-players-detection-1/README.robotflow	Robotflow dataset provenance and license details.

Function and Method Inventory

Symbol	File	Parameters	Called By
main	main.py	-	No direct local match
Tracker.__init__	trackers/tracker.py	self, model_path	No direct local match
Tracker.add_position_to_tracks	trackers/tracker.py	self, tracks	main
Tracker.interpolate_ball_positions	trackers/tracker.py	self, ball_positions	main
Tracker.detect_frames	trackers/tracker.py	self, frames	Tracker.get_object_tracks
Tracker.get_object_tracks	trackers/tracker.py	self, frames, read_from_stub, stub_path	main
Tracker.draw_ellipse	trackers/tracker.py	self, frame, bbox, color, track_id	Tracker.draw_annotations
Tracker.draw_triangle	trackers/tracker.py	self, frame, bbox, color	Tracker.draw_annotations
Tracker.draw_team_ball_control	trackers/tracker.py	self, frame, frame_num, team_ball_control	Tracker.draw_annotations
Tracker.draw_annotations	trackers/tracker.py	self, video_frames, tracks, team_ball_control	
TeamAssigner.__init__	team_assigner/team_assigner.py	self	No direct local match
TeamAssigner.get_clustering_model	team_assigner/team_assigner.py	self, image	TeamAssigner.get_player_color
TeamAssigner.get_player_color	team_assigner/team_assigner.py	self, frame, bbox	TeamAssigner.assign_team_color, TeamAssigner.get_player_team
TeamAssigner.assign_team_color	team_assigner/team_assigner.py	self, frame, player_detections	main
TeamAssigner.get_player_team	team_assigner/team_assigner.py	self, frame, player_bbox, player_id	main
PlayerBallAssigner.__init__	player_ball_assigner/player_ball_assigner.py	self	No direct local match
PlayerBallAssigner.assign_ball_to_player	player_ball_assigner/player_ball_assigner.py	self, player, player_ball_bbox	main
CameraMovementEstimator.__init__	camera_movement_estimator/camera_movement_estimator.py	self, frame	No direct local match
CameraMovementEstimator.get_camera_movement	camera_movement_estimator/camera_movement_estimator.py	self, gray_frame	CameraMovementEstimator.get_camera_movement
CameraMovementEstimator.add_adjusted_position_to_tracks	camera_movement_estimator/camera_movement_estimator.py	self, tracks, camera_movement, main_frame	
CameraMovementEstimator.get_camera_movement	camera_movement_estimator/camera_movement_estimator.py	self, frame, read_from_stub, stub_path	main
CameraMovementEstimator.draw_camera_movement	camera_movement_estimator/camera_movement_estimator.py	self, frame, camera_movement, main_frame	
SpeedAndDistanceEstimator.__init__	speed_and_distance_estimator/speed_and_distance_estimator.py	self	No direct local match
SpeedAndDistanceEstimator.add_speed_and_distance_to_tracks	speed_and_distance_estimator/speed_and_distance_estimator.py	self, tracks	main
SpeedAndDistanceEstimator.draw_speed_and_distance	speed_and_distance_estimator/speed_and_distance_estimator.py	self, frame, tracks	main
ViewTransformer.__init__	view_transformer/view_transformer.py	self	No direct local match
ViewTransformer.transform_point	view_transformer/view_transformer.py	self, point	ViewTransformer.add_transformed_position_to_tracks
ViewTransformer.add_transformed_position_to_tracks	view_transformer/view_transformer.py	self, tracks	main
get_center_of_bbox	utils/bbox_utils.py	bbox	PlayerBallAssigner.assign_ball_to_player, Tracker.add_position_to_tracks
get_bbox_width	utils/bbox_utils.py	bbox	Tracker.draw_ellipse
measure_distance	utils/bbox_utils.py	p1, p2	CameraMovementEstimator.get_camera_movement, SpeedAndDistanceEstimator.add_speed_and_distance_to_tracks
measure_xy_distance	utils/bbox_utils.py	p1, p2	CameraMovementEstimator.get_camera_movement
get_foot_position	utils/bbox_utils.py	bbox	SpeedAndDistanceEstimator.draw_speed_and_distance
read_video	utils/video_utils.py	path, resize_width, max_frames	main
save_video	utils/video_utils.py	output_video_frames, output_video_path	main

Libraries and Dependencies

Dependency	Where/Why used
ultralytics	Detection inference and notebook-based training
supervision	ByteTrack and detection conversion
opencv-python	Video I/O, drawing, optical flow, transforms
numpy	Arrays and geometry
pandas	Ball interpolation
scikit-learn	KMeans team assignment